# Modern Algebra

Assignment 11

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## 11 Finite Abelien Groups

## Problem 2

We want to find an integer $n$ such that it is the least integer which we can find three different partitions for the powers of the primes in its prime factorization. To find such an integer, choose the least prime number, which is 2. Since 3 has three partitions, we have three options:

- partition 3: $\mathbb{Z}_{2^3}$
- partition 2+1: $\mathbb{Z}_{2^2} \oplus \mathbb{Z}_2$
- partition 1+1+1: $\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$

## Problem 5

By the Fundamental Theorem of Finite Abelian Groups, observe that an abelian group of order 45 is isomorphic to $G_1 \cong \mathbb{Z}_5 \oplus \mathbb{Z}_{3^2}$ or $G_2 \cong \mathbb{Z}_5 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_3$. We know that the maximum order of an element in $G_1, G_2$ is $\text{lcm}(9,5) = 45, \text{lcm}(3,5) = 15$ respectively. Since 15 divides both $15, 45$. Thus, the first statement is true. For the second statement, since $9 \nmid 15$, not all abelian groups of order 45 has an element of order 9.

# Problem 10

We know that the prime factorization of 360 is $5 \cdot 3^2 \cdot 2^3$. By the Fundamental Theorem of Finite Abelian Groups, any abelian group of order 360 is isomorphic to one of the following partitions:

- $\mathbb{Z}_5 \oplus \mathbb{Z}_{3^2} \oplus \mathbb{Z}_{2^3}$
- $\mathbb{Z}_5 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_{2^3}$
- $\mathbb{Z}_5 \oplus \mathbb{Z}_{3^2} \oplus \mathbb{Z}_{2^2} \oplus \mathbb{Z}_2$
- $\mathbb{Z}_5 \oplus \mathbb{Z}_{3^2} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$
- $\mathbb{Z}_5 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_{2^2} \oplus \mathbb{Z}_2$
- $\mathbb{Z}_5 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$

# Problem 22

*Proof.* By the Fundamental Theorem of Finite Abelian Groups, we have a sequence of primes $p_1, p_2, p_3, \cdots, p_k$ and a sequence of integers $n_1, n_2, n_3, \cdots, n_k$ such that

$$G \cong \mathbb{Z}_{p_1^{n_1}} \oplus \mathbb{Z}_{p_2^{n_2}} \oplus \mathbb{Z}_{p_3^{n_3}} \oplus \cdots \oplus \mathbb{Z}_{p_k^{n_k}}$$

To prove that $G$ is cyclic, we must show that the primes $p_1, p_2, p_3, \cdots, p_k$ are all distinct. Assume for the sake of contradiction that the group is not cyclic, so we have $p_1 = p_2 = p$. We find $H_1 = \mathbb{Z}_{p^{n_1}} \oplus \{0\} \oplus \{0\} \oplus \cdots \oplus \{0\}, H_2 = \{0\} \oplus \mathbb{Z}_{p^{n_2}} \oplus \{0\} \oplus \cdots \oplus \{0\}$ are both cyclic subgroups in $G$. Clearly, each of $H_1, H_2$ has a subgroup of order $p$. Thus, we found two distinct subgroups of order $p$ contradicting that $G$ has exactly one subgroup for each divisor of $|G|$. Hence, $G$ is cyclic. ■

# Problem 31

By the Fundamental Theorem of Finite Abelian Groups, we know that $G$ is isomorphic to one of five following groups:

1. $\mathbb{Z}_{2^4}$
2. $\mathbb{Z}_{2^3} \oplus \mathbb{Z}_2$
3. $\mathbb{Z}_{2^2} \oplus \mathbb{Z}_{2^2}$
4. $\mathbb{Z}_{2^2} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$
5. $\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$

Clearly, we eliminate the fifth group as it does not contain elements of order 4. We eliminate groups 1, 2, and 4 because in these cases, the squares of all elements of order 4 are identical:

- In Group 1 ($\mathbb{Z}_{16}$), the elements of order 4 are 4 and 12. Their squares are $2(4) = 8$ and $2(12) = 24 \equiv 8$. Thus $a^2 = b^2$.
- In Group 2 ($\mathbb{Z}_8 \oplus \mathbb{Z}_2$), any element $g = (x, y)$ of order 4 must have $x \in \{2, 6\}$. The square is $2g = (2x, 2y) = (4, 0)$. Thus $a^2 = b^2$.
- In Group 4 ($\mathbb{Z}_4 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$), any element $g = (x, y, z)$ of order 4 must have $x \in \{1, 3\}$. The square is $2g = (2x, 2y, 2z) = (2, 0, 0)$. Thus $a^2 = b^2$.

Observe the third case $\mathbb{Z}_{2^2} \oplus \mathbb{Z}_{2^2}$, consider $a = (1, 0), b = (0, 1)$ elements of order 4 in $\mathbb{Z}_{2^2} \oplus \mathbb{Z}_{2^2}$. See that $(0, 1) + (0, 1) = (0, 2)$ and $(1, 0) + (1, 0) = (2, 0)$. Thus, $a^2 \neq b^2$. Hence, $G \cong \mathbb{Z}_{2^2} \oplus \mathbb{Z}_{2^2}$.